

**Arabinose-Dependent Kill Switches in *Pseudomonas putida* for Safe Oil Spill
Bioremediation**

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Abstract:

The 1989 Exxon Valdez oil spill that occurred in Prince William Sound, Alaska, highlighted the devastating impacts of oceanic oil spills and the urgent need for effective cleanup methods. In response, scientists have used genetically engineered bacteria as a solution; however, their uncontrolled spread raises critical biosafety concerns. This study investigates *Pseudomonas putida*, a non-pathogenic, naturally oil-degrading bacterium, modified with an arabinose-dependent kill switch. This synthetic circuit ensures that the bacteria survive only when arabinose is present, making it possible to terminate them after cleanup by removing the nutrient. Our experiment will test the effectiveness and stability of this kill switch, comparing bacterial survival and spread across four conditions: engineered and unmodified bacteria, each with and without arabinose. We expect that engineered bacteria will die without arabinose and remain stable with it, while control bacteria will survive in both. This approach aims to offer a controllable, eco-safe bioremediation tool.

Introduction:

When the Exxon Valdez oil spill occurred, over 10 million gallons of crude oil were released into the nearby waters, triggering catastrophic environmental damage (NOAA, n.d.). Traditional physical and chemical cleanup methods proved insufficient and ecologically harmful. Recently, genetically modified bacteria like *Pseudomonas putida* have emerged as solutions due to their ability to degrade hydrocarbons (Rafeeq et al., 2023). However, their release into open ecosystems has raised valid concerns about containment and mutation. Scientists have incorporated “kill switches,” specially designed genetic controls that stop bacteria from surviving once they are outside specific conditions.

A nutrient-based kill switch offers a promising solution by engineering bacteria to rely on a synthetic nutrient that does not naturally or often occur in the contaminated environment. The absence of this nutrient prevents bacterial survival, thereby limiting their spread and reducing ecological risks. If modified bacteria escape into the wild, they could reproduce uncontrollably, disrupt natural microbial communities, or transfer the engineered genes to native organisms.

Although previous studies have explored kill switches in bacteria, most have been limited to genetic-level analysis or ideal lab conditions (Lawrence Livermore National Laboratory, n.d.). Our study aims to bridge this gap by testing a kill switch in a controlled environment while measuring real-world indicators like survival, spread, and mutation. The core of our project lies in engineering *P. putida* (*Pseudomonas putida*) to survive only in the presence of arabinose, a sugar uncommonly found in natural environments. This allows us to simulate both permissive and restrictive environments and test whether the bacteria behave as intended.

Hypothesis:

If *P. putida* is engineered with an arabinose-dependent kill switch, then it will only survive and degrade oil when arabinose is present. If it is absent, the kill switch will activate and lead to bacterial death, minimal spread, and reduced mutation risk.

Methods:

We propose a comparative experiment with 4 groups to prove the viability of arabinose-based kill switches in bioengineered oil-degrading bacteria.

Independent Variable: presence of kill switch

Dependent Variable: bacteria survival and spread

The bacteria we will use for this experiment are *Pseudomonas putida* (*P. putida*), a popular bioremediation bacterium that breaks down hydrocarbons in oil spills. Multiple studies support its potential for kill switch genetic modification (Raghavan & Vivekanandan, 1999). The switch is based on the nutrient arabinose, which is likely to be compatible with *P. putida*.

By connecting the gene switch PBAD Promoter (which turns the DNA in the bacteria on/off based on arabinose presence), the nutrient is outside the cell, and can enter and turn the bacteria's oil-degrading properties on. Our experiment will include two main groups: an experimental group consisting of bioengineered *P. putida* containing the arabinose kill switch, designed to die off once bioremediation has been successful and the synthetic nutrient is removed; and a control group made up of identical bioengineered *P. putida* without any switch, expected to survive after bioremediation and potentially spread in natural conditions. We will simulate two environments for each group: one of the synthetic nutrients that the kill switch relies on, and one without the nutrient.

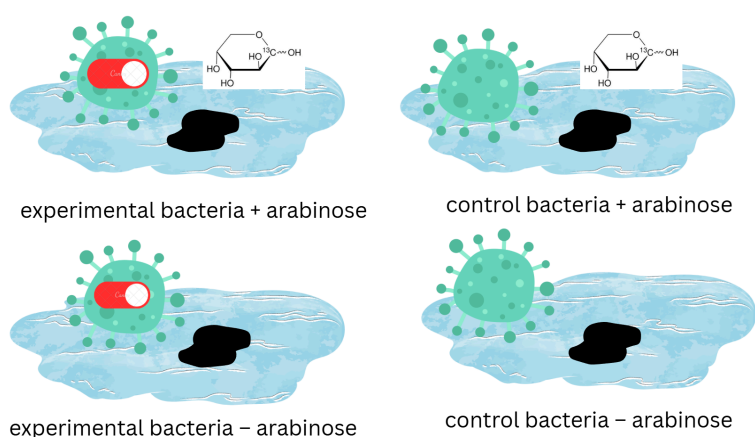


Figure 1: The four conditions: experimental bacteria + arabinose (expected to be effective,

stable, and survive), experimental bacteria – arabinose (expected to die), control bacteria + arabinose (normal function), and control bacteria – arabinose (normal function). Each contains an oil spill that is expected to be remediated.

Constants such as oil level, type of bacteria, temperature, exposure time, pH levels will be controlled across all conditions.

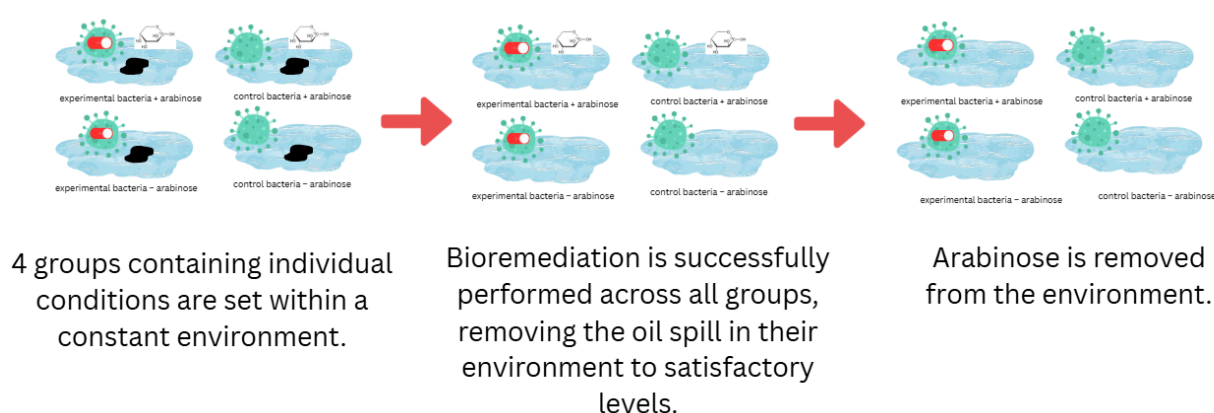


Figure 2: Steps for the first part of the procedure. Measurements are observed after 15 days. Efficiency of oil degradation will be measured during bioremediation (step 2).

We will measure and observe survival rates using Colony Forming Unit/ml over time, reproductive rates through optical density using light, the efficiency of oil degradation, the spatial and temporal spread and mutation (frequency = # of escaped colonies divided by # of bacteria. For the experimental group, mutants will be identified as the ones surviving without the nutrient, indicating that the kill switch has failed.

Our experimental design is straightforward and easy to conduct, while offering clear and comprehensive evidence for the hypothesis. Ultimately, the study will contribute valuable data

toward safer applications of bioengineered bacteria for environmental cleanup.

Conclusion:

Based on our proposed experiment, it is predicted that the genetically modified *P. putida* containing the arabinose kill switch will fail to survive or spread once the nutrient is removed, while effectively performing bioremediation when the arabinose is present. These results will demonstrate that kill switches are a reliable solution for controlling bioengineered bacteria. What makes the study novel is that it tests the kill switch functions in a controlled environment so variables can be controlled and measured carefully. Unlike other studies that focus solely on genetic functions, we observe additional key outcomes, creating a more complete picture of kill switch behaviour in practice. However, it is limited to a lab environment and does not account for real-world variables like different temperatures or natural microbial interactions. In the future, research can explore multilayered switches or test under complex lab simulations. Overall, our proposal will support safer synthetic biology applications in the future for environmental cleanup.

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